

```
from google.colab import files
uploaded = files.upload()
```

mQtsrFlnID...qgAO9Y.jpg

mQtsrFlnIDjGwmdGSF3E5VTUtDhQfoQvLxiHVwpLHXIXWZWREtyMpxEdAmgW4ygP5q4sY57RFYmJqxqxIUl_lpumqUCkaA78b2s9wyGGhU
KA1inwq5XjlgRhGAFwZK9Y7AXLXnxFSvcZqgAO9Y.jpg(image/jpeg) - 18911 bytes, last modified: 7/22/2026 - 100% done

Saving mQtsrFlnIDjGwmdGSF3E5VTUtDhQfoQvLxiHVwpLHXIXWZWREtyMpxEdAmgW4ygP5q4sY57RFYmJqxqxIUl_lpumqUCkaA78b2s9wyGGhU0FCqPn5

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os

# Get the filename of the uploaded image
image_filename = list(uploaded.keys())[0]

# Read image using the correct filename
img = cv2.imread(os.path.join('/content/', image_filename))

# Check if image was loaded successfully
if img is None:
    print(f"Error: Could not load image '{image_filename}'. Please check the file path and ensure the
else:
    # Convert to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Create kernel
    kernel = np.ones((5,5), np.uint8)

    # Apply dilation
    dilated = cv2.dilate(gray, kernel, iterations=1)

    # Display images
    titles = ['Original (Grayscale)', 'Dilated']
```

```
images = [gray, dilated]

plt.figure(figsize=(10, 5))
for i in range(2):
    plt.subplot(1,2,i+1)
    plt.imshow(images[i], cmap='gray')
    plt.title(titles[i])
    plt.axis('off')

plt.tight_layout()
plt.show()
```

Original (Grayscale)

Original Image



Dilated



